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Learn
Build
Grow

SNOWFLAKE

WITH REAL-WORLD CONCEPTS



NOTES

BEGINNER TO ADVANCED

SWIPE TO NEXT



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Snowflake - The Big Picture

A modern, cloud-native data platform for the data-driven world.

① What is Snowflake?

Snowflake is a **cloud-native data platform** that lets you store, process, analyze and securely share data at scale. It runs on major cloud providers (AWS, Azure, GCP) and is designed for modern data workloads.



Key Takeaway

Snowflake = Store + Process + Analyze + Share data in the cloud, without managing infrastructure. ✓

② Why Does Snowflake Exist?

- Traditional data systems are **complex, costly and hard to scale**.
- Data is **growing fast** and organizations need a platform that can handle large volumes of data easily.
- Snowflake was built to **simplify data infrastructure** and make analytics easy, fast and scalable.

③ Problems Snowflake Solves

- ✓ **Eliminates** infrastructure management
- ✓ **Handles large and growing** data volumes
- ✓ Provides **high scalability and performance**
- ✓ Enables **secure data sharing**
- ✓ Reduces **cost and complexity**
- ✓ Helps teams focus on analytics and innovation, not on managing hardware

④ Why Organizations Use Snowflake

- ✓ **On-demand scalability**
- ✓ **Pay only** for what you use
- ✓ Supports diverse data **workloads**
- ✓ **High availability and reliability**
- ✓ Easy **data sharing** and collaboration
- ✓ Works across **multiple cloud providers**
- ✓ Helps accelerate **data-driven decisions**

⑤ Where Snowflake Fits?

Snowflake sits at the center of **modern data architecture**, connecting data from multiple sources and enabling **analytics, machine learning and business intelligence**. It acts as a unified platform for all your data needs in the cloud.

⑥ Traditional Data Platform vs Snowflake

Traditional Data Platform	Snowflake (Cloud Data Platform)
 • Buy and manage hardware • Long setup time • Limited scalability • High upfront cost • Manual maintenance • Complex to manage	 • No hardware to manage • On-demand resources • Scales easily • Pay as you go • Fully managed by Snowflake • Focus on analytics & innovation



Remember

Snowflake is not just a database, it's a complete cloud data **platform** built for the future. ✓

Simpler Data. Bigger Possibilities.

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Snowflake Architecture

A simple, powerful and unique cloud data platform architecture.

1 The 3 Core Layers

Snowflake has a simple and powerful architecture with three separate layers that work together:

1. **Cloud Services** → Manages and coordinates
2. **Compute** → Runs your queries and workloads
3. **Storage** → Stores your data securely

3 Compute Layer (Virtual Warehouses)

- Provides **independent compute resources** to run queries and workloads.
- Multiple virtual warehouses can run at the same time, independent of each other.
- Compute is **separate from storage**, so you can scale it based on your needs.

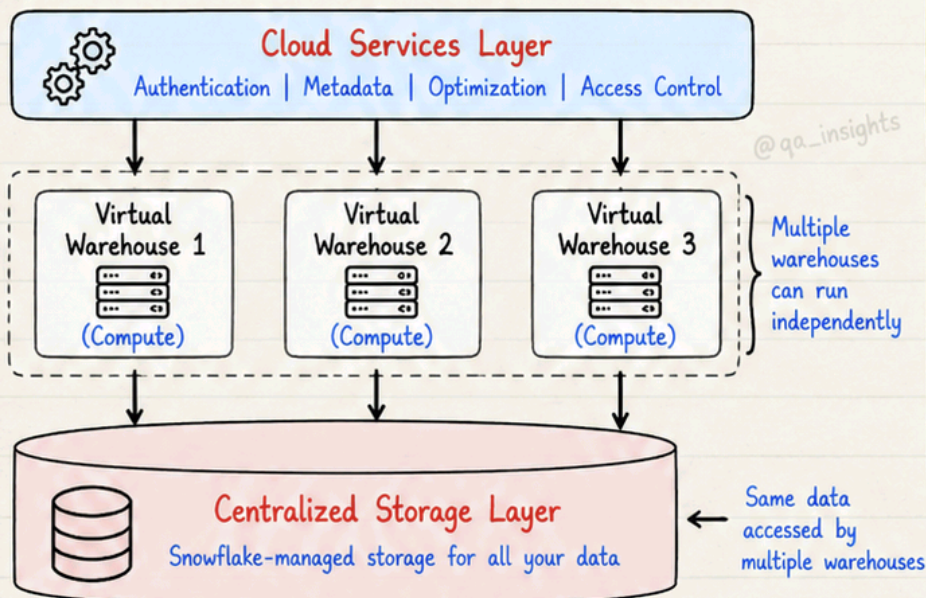
2 Cloud Services Layer

- Coordinates user **authentication**, **metadata management**, **query optimization** and **access control**.
- Acts as the brain of Snowflake and connects compute and storage.
- Automatically manages infrastructure tasks so you can focus on your data.

4 Storage Layer

- A centralized, **Snowflake-managed** storage layer.
- Stores all your data (structured, semi-structured and unstructured).
- Data is stored **separately from compute**, allowing independent scaling.
- Highly durable and secure.

5 Snowflake Architecture (High-Level View)



Key Takeaway

Snowflake **separates storage from compute** — so each can scale independently. ✓

Why This Architecture Matters?

- ✓ Storage and compute scale independently
- ✓ Workloads can be isolated
- ✓ Reduced infrastructure management
- ✓ Helps organizations focus on data and analytics



Remember

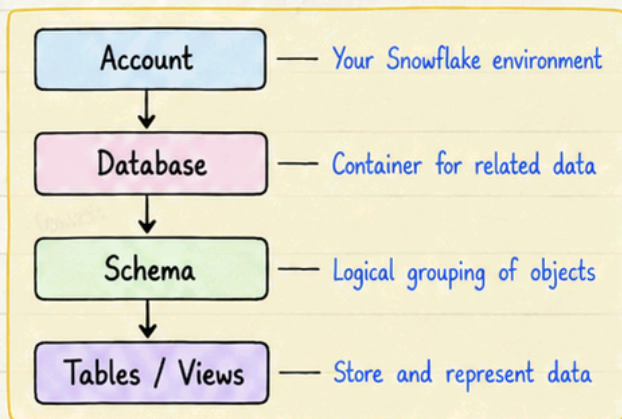
Virtual Warehouse = compute, **not** storage. ✓

Databases, Schemas & Tables

Organizing data in Snowflake using a simple and logical hierarchy.

1 Snowflake Object Hierarchy

Snowflake organizes data using a **hierarchical structure**. Objects are created within an account and follow this order:



2 Key Objects in Snowflake

Object	Purpose
Account	Top-level container for your organization.
Database	Holds related schemas and data objects.
Schema	Logical container for tables, views and more.
Table	Stores structured data (rows and columns).
View	Virtual table based on a query.
Stage	Used to load and unload data files.
File Format	Defines the structure of data files.
Sequence	Generates sequential numbers.
Function	Reusable SQL logic.

3 What are Databases?

- A database is a **container** for related data.
- You can have **multiple databases** in an account.
- Helps in organizing data by domain, project or business unit.

4 What are Schemas?

- A schema is a **logical grouping** of data objects.
- A database can have **multiple schemas**.
- Helps in organizing objects like tables, views, functions etc.

5 What are Tables and Views?

- A table stores **actual data** in rows and columns.
- A view is a **virtual table** based on a SQL query.
- Views do not store data, they only store the query.

6 Why This Structure Matters?

- ✓ Keeps data **well organized**
- ✓ Improves **manageability** and access control
- ✓ Supports **multiple teams** and use cases
- ✓ Makes it **easy to find** and work with data

Best Practice

- ✓ Use **meaningful names** for databases and schemas.
- Organize data based on business domains.
- Follow consistent naming conventions.
- Use schemas to separate environments (dev, test, prod).

Remember

- 💡 **Account** → **Database** → **Schema** → **Tables/Views**
- A schema cannot exist without a database.
- Views are **queries**, not actual data.
- This structure helps in better data organization and governance.

Virtual Warehouses in Snowflake

Compute power to run your queries, separate from storage.

1 What is a Virtual Warehouse?

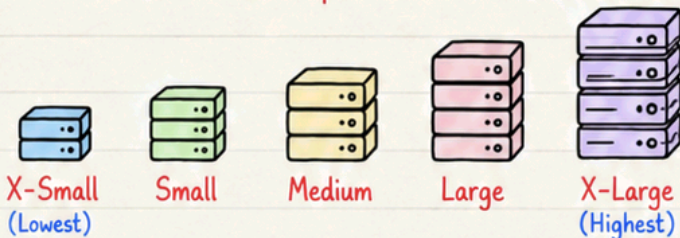
- A **virtual warehouse (VW)** is a cluster of **compute resources** used to run queries and data processing tasks in Snowflake.
- It provides the **compute power** (CPU, memory) needed to execute your SQL queries.
- Multiple warehouses can run at the same time, independent of each other.

2 Why Do We Need Compute?

- Data is stored in Snowflake's centralized storage, but **compute** is needed to:
 - ✓ Run SQL queries
 - ✓ Transform and process data
 - ✓ Support multiple users and workloads
 - ✓ Scale based on demand
 - ✓ Keep storage and compute separate

3 Warehouse Sizes

- Snowflake provides different warehouse sizes to handle varying workloads.
- You can choose the right size based on your performance and cost needs.
- You can also **scale up or down** as needed.



4 Scaling, Suspend and Resume

- Scale Up:** Increase the warehouse size to handle more workload.
- Scale Down:** Decrease the size to save cost.
- Auto-Suspend:** Automatically stops the warehouse after a period of inactivity.
- Auto-Resume:** Automatically starts the warehouse when a new query arrives.

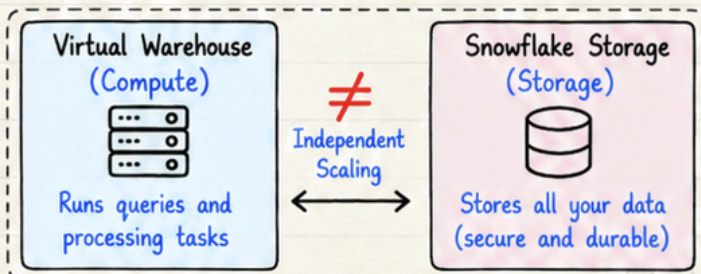
Key Takeaway



Virtual warehouses can be scaled, suspended and resumed without affecting your data in storage. ✓

5 Compute vs Storage (Separation)

- In Snowflake, **compute** and **storage** are completely separate and can scale independently.
- This allows better performance, flexibility and cost efficiency.



6 Things to Remember

- ✓ You can have **multiple** warehouses.
- ✓ Choose the right size for your workload.
- ✓ Use **auto-suspend** to save cost.
- ✓ Compute and storage scale independently.
- ✓ You can resize a warehouse anytime.

Common Mistake



A virtual warehouse does **NOT** store data. It only **provides compute** to run queries. Your data always stays in Snowflake storage.

Data Loading & Ingestion in Snowflake

Get your data into Snowflake from various sources.

1 What is Data Ingestion?

- Data ingestion is the process of loading data from **external sources** into Snowflake.
- Snowflake supports both **batch** and **continuous** (streaming) ingestion.
- Data is first loaded into a **stage** and then moved into **tables**.

2 Stages in Snowflake

- Internal Stage:** Stored inside your Snowflake account.
- External Stage:** Points to cloud storage like Amazon S3, Azure Blob or Google Cloud Storage.
- Stages temporarily hold data files before loading into tables.

3 File Formats

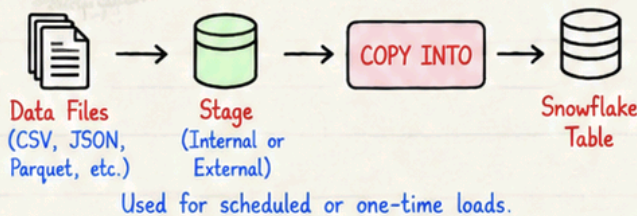
- A **file format** defines the structure of your data files.
- Common formats: **CSV, JSON, Parquet, Avro, ORC, XML**.
- Helps Snowflake correctly parse the data while loading.

4 How to Load Data?

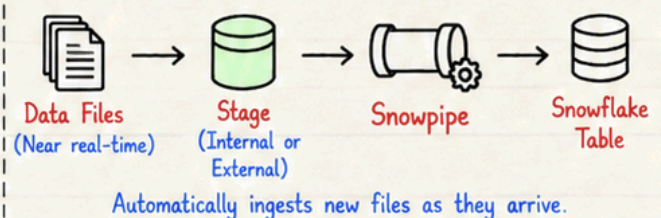
- Use **COPY INTO** command to load data from a stage into a table.
- Snowflake automatically handles file parsing, compression and parallel loading.
- You can load data from **internal** or **external** stages using **COPY INTO**.

5 Data Loading Methods

Batch Loading (Manual / Scheduled)



Continuous Ingestion (Snowpipe)



6 Batch vs Continuous Ingestion

Feature	Batch Loading	Snowpipe (Continuous)
How it works	Manual or scheduled COPY INTO	Automatically loads new files
Use case	Periodic / large loads	Real-time / streaming
Latency	Minutes / Hours	Near real-time (minutes)
Setup	Simple	Requires configuration
Cost	Pay for compute used	Pay for compute (auto)



Key Takeaway

- Use stages to store your data files.
- Use **file formats** to define structure.
- Use **COPY INTO** for batch loading.
- Use **Snowpipe** for continuous ingestion.
- Choose the right method based on your business needs.



Common Mistake

- Trying to load files directly into tables without using a stage.
- Using the wrong file format.
- Expecting real-time ingestion without Snowpipe.

SQL & Data Transformation in Snowflake

Query, transform and prepare your data for analytics.

1 Querying Data in Snowflake

- Use **SQL** to query data stored in Snowflake.
- Supports standard SQL with high performance.
- You can query data from **tables**, **views** and **external/internal stages**.
- Works with structured, semi-structured and unstructured data.

2 Common SQL Operations

Operation	Purpose
SELECT	Retrieve data from a table
WHERE	Filter data based on conditions
JOIN	Combine data from multiple tables
GROUP BY	Aggregate data
ORDER BY	Sort the results
CTEs (WITH)	Create temporary result sets
MERGE	Insert, update or delete data
Subqueries	Use query inside another query

3 Example SQL Queries

1. Select & Filter

```
SELECT *
FROM sales
WHERE region = 'US';
```

2. Join

```
SELECT c.customer_name, o.order_id
FROM customers c
JOIN orders o
ON c.id = o.customer_id;
```

3. Group By

```
SELECT region, COUNT(*) as total
FROM sales
GROUP BY region;
```

4. CTE (WITH)

```
WITH recent_sales AS (
  SELECT * FROM sales
  WHERE order_date >= '2024-01-01'
)
SELECT * FROM recent_sales;
```

4 Data Transformation

- Transform raw data into clean, analysis-ready datasets using SQL.
- Perform filtering, joining, aggregations and calculations.
- Create **views** or new tables with transformed data.
- Use **MERGE** for upserts (insert/update/delete).
- Helps prepare data for reporting and analytics.

5 Semi-Structured Data Support

- Snowflake can query semi-structured data like **JSON**, **Avro**, **Parquet** and **XML**.
- Use functions like :
 - **VARIANT** data type
 - Parse JSON using : | : and **FLATTEN()**.
 - Extract and transform nested data easily.

6 SQL Transformation Workflow



Key Takeaway

- ✓ Use SQL to query and transform data.
- ✓ Snowflake supports structured and semi-structured data.
- ✓ CTEs and **MERGE** help in data preparation.
- ✓ Transform data for better analytics and business insights.



Common Mistake

- Not using filters (scans more data).
- Incorrectly parsing semi-structured data.
- Not using CTEs for complex queries.

Snowflake Data Sharing & Collaboration

Share data securely without copying it.

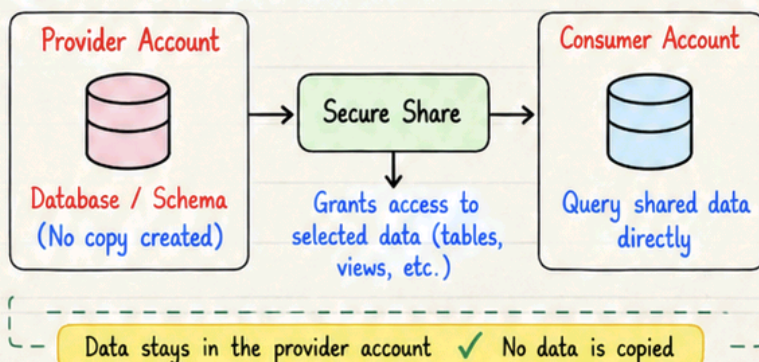
1 What is Secure Data Sharing?

- **Secure Data Sharing** allows you to share live data with other Snowflake accounts without creating a copy of the data.
- The shared data remains in the **provider's Snowflake account**.
- Consumers can query the data directly, following the permissions set by the provider.

2 Key Components

Component	Purpose
Database	Container for schemas and objects to be shared.
Share	Logical object used to share data with other accounts.
Consumer	The account that receives and accesses the shared data.
Reader Account	A special type of Snowflake account to consume shared data (optional).
Listing (Marketplace)	Make shared data available publicly via Snowflake Marketplace.

3 How Data Sharing Works?



4 Types of Sharing

- **Account-to-Account Sharing** → Directly share data with specific Snowflake accounts.
- **Reader Account Sharing** → Share data with a reader account (no storage, only access).
- **Marketplace Sharing** → List your data on Snowflake Marketplace for a wider audience.
- **Live Data Access** → Consumers always access the latest data, no need to copy or move.

5 Why It's Useful?

- ✓ Share data **securely** across teams, partners or customers.
- ✓ No need to create and manage data copies.
- ✓ **Real-time** access to the latest data.
- ✓ Maintain full **control** over permissions.
- ✓ Reduces **storage costs** and data movement.
- ✓ Enables easy collaboration and data monetization.

6 Real-World Examples



Key Takeaway

- Snowflake enables secure, live data sharing without copying the data.
- Use **shares** to control what data is shared.
- Great for collaboration, analytics and monetization.

Common Mistake



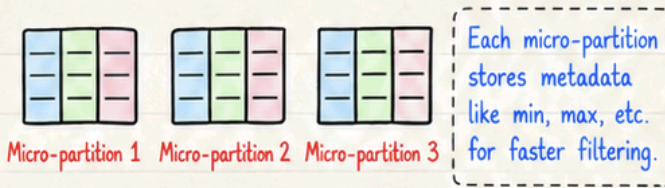
- Thinking that shared data is copied to the consumer account.
- Over-sharing data without proper access control.
- Not using the right sharing method for your use case.

Performance & Cost Optimization in Snowflake

Run faster queries and optimize costs with the right practices.

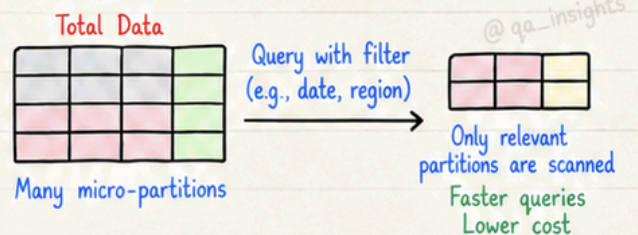
1 What are Micro-partitions?

- Snowflake automatically divides data into **micro-partitions** (small, columnar storage units).
- Each micro-partition contains **metadata** (min, max, etc.) for efficient query processing.
- No manual partitioning is required.



2 Partition Pruning

- Snowflake scans only the relevant micro-partitions that match your query filters.
- This is called **partition pruning**.
- Less data scanned = faster performance and lower cost.



3 Clustering (Optional)

- Clustering** organizes related data together in micro-partitions.
- Useful for large tables with frequent filters on specific columns.
- Helps improve query performance by increasing the effectiveness of partition pruning.

4 Caching in Snowflake

- Snowflake automatically caches frequently accessed data and query results.
- Types of caching:
 - Data cache** → Frequently used data.
 - Result cache** → Previous query results.
 - Metadata cache** → Metadata about tables, partitions, etc.
- Caching helps improve performance and reduce cost.

5 Warehouse Sizing

- Choose the right warehouse size based on your workload and performance needs.
- You can **scale up or down** as needed.



6 Auto-Suspend and Auto-Resume

- Auto-Suspend**: Automatically stops the warehouse after a period of inactivity.
- Auto-Resume**: Automatically starts the warehouse when a new query arrives.
- Helps reduce unnecessary compute costs.

Remember

💡 Compute stops when not in use. You only pay for the time it runs.

7 Cost Control Principles

Right Size Use appropriate warehouse size	Auto-Suspend Turn on auto-suspend	Optimize Queries Use filters and avoid full table scans	Monitor Usage Track usage and control costs

Common Mistake

- ✗ Using always-on warehouses.
- ✗ Running large queries without filters.
- ✗ Choosing oversized warehouses.
- ✗ Not enabling auto-suspend.
- ✗ Ignoring query performance and scanned data.

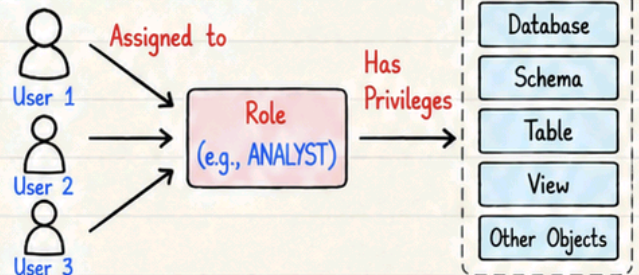
Security, Governance & Advanced Features in Snowflake

Keep your data secure, compliant and available.

1 Role-Based Access Control (RBAC)

- Snowflake uses **roles** to control access to databases, schemas, tables and other objects.
- A role is granted **privileges** and assigned to users or other roles.
- Follow the principle of **least privilege**.

2 Roles, Privileges and Users



3 Data Masking

- Data masking** hides sensitive data (e.g., PII) from unauthorized users.
- Uses **masking policies** to return obfuscated data.
- Helps protect sensitive information while allowing data usage.

Example

Original Data			Masking Policy	Masked Data		
ID	Name	Email		ID	Name	Email
1	John Doe	john@example.com	→	1	John Doe	j***@example.com

4 Row Access Policies

- Row access policies** filter rows based on user roles or attributes.
- Different users can see different data from the same table.
- Useful for multi-tenant or department-level data access.

Example

User Role	Data Visible
SALES	Only sales data
HR	Only HR data
ANALYST	All data (if permitted)

5 Time Travel

- Snowflake automatically keeps **historical** data for a configurable period (e.g., 1-90 days).
- You can query past data or restore objects.
- Useful for recovery, auditing and accidental deletes.

6 Fail-safe

- After the Time Travel period, dropped data is retained in **Fail-safe** (7 days).
- Helps protect against accidental or malicious data loss.

7 Zero-Copy Cloning

- Create an **instant**, space-efficient copy of databases, schemas or tables.
- No data is physically copied.**
- Useful for testing, development and backups.

8 Governance & Compliance

- Snowflake provides features to help with **data governance, auditing and compliance**.
- Track access using **ACCOUNT_USAGE** views.
- Helps meet regulatory requirements (e.g., GDPR, HIPAA).



Key Takeaway

- ✓ Use roles and least privilege.
- ✓ Protect sensitive data with masking and row access policies.
- Use Time Travel and Fail-safe for data protection.
- Leverage cloning for non-production use cases.

Common Mistake



- Giving too many privileges.
- Not using masking for sensitive data.
- Relying only on Time Travel (forgetting Fail-safe exists).
- Not following governance and compliance best practices.

End-to-End Snowflake Workflow

From data to insights — A complete real-world pipeline.

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1 What is an End-to-End Workflow?

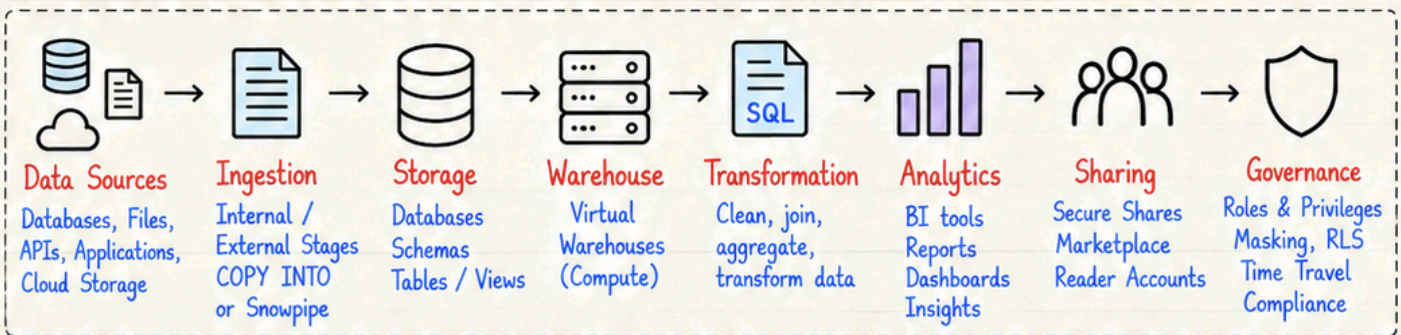
- A complete data pipeline from **source systems** to analytics and sharing using Snowflake.
- Helps organizations **ingest, store, process, analyze** and securely **share** data at scale.
- Brings together all the key Snowflake concepts we learned in this series.

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2 Key Steps in the Workflow

1. **Data Sources** → Collect data from various systems.
2. **Ingestion** → Load data using stages, **COPY INTO** or Snowpipe.
3. **Storage** → Data is stored in Snowflake (databases, schemas, tables).
4. **Warehouse** → Compute with virtual warehouses.
5. **Transformation** → Use SQL to clean, join and transform data.
6. **Analytics** → Use BI tools for reporting and insights.
7. **Sharing** → Share data securely with other accounts.
8. **Governance** → Manage access, security and compliance.

3 End-to-End Snowflake Pipeline



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A scalable, secure and unified data platform with Snowflake.

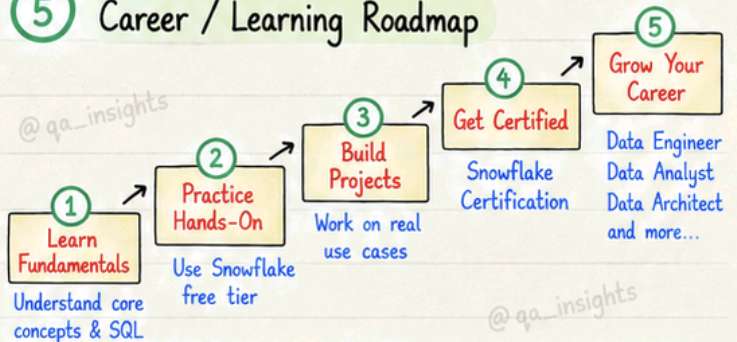
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4 Real-World Use Cases

- ✓ **Analyze** customer data across multiple systems.
- ✓ **Build** real-time analytics pipelines.
- ✓ **Share** data with partners without copying.
- ✓ Create a single source of truth for the organization.
- ✓ Power data-driven decision making.

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5 Career / Learning Roadmap



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6 Key Takeaways



- ✓ Snowflake simplifies the entire data lifecycle.
- ✓ Compute and storage are separate and independent.
- ✓ It enables secure sharing without copying data.
- ✓ Follow best practices for performance, cost and governance.
- ✓ Great opportunities for career growth in the data cloud space.

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Common Mistake



- Not using the right ingestion method.
- Running large queries without optimization.
- Keeping warehouses running when not in use.
- Ignoring security and governance.
- Not exploring real-world use cases.

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